



MVJ COLLEGE OF ENGINEERING BANGALORE-560067

(An Autonomous Institute)

A MINI PROJECT REPORT

ON

“PERSONAL EXPENSE TRACKER”

*Submitted in practical fulfillment for the award of degree of Bachelor of
Engineering in*

INFORMATION SCIENCE AND ENGINEERING

SUBMITTED BY

AKSHAY A GOWDA

1MJ22IS009

CHETHAN SDN

1MJ22IS022

DEEKSHITH H A

1MJ22IS023

DHIRAJ A ACHARYA

1MJ22IS025

UNDER THE GUIDANCE OF

Prof. GAYATHRI

Assistant Professor of ISE Department



MVJ College of Engineering, Bangalore
(An Autonomous Institute)

Affiliated to VTU, Belgavi, Approved by AICTE, New Delhi, Recognized By
UGC with 2(f)&12(B), Accredited by NBA & NAAC 2024-25



MVJ COLLEGE OF ENGINEERING, BENGALURU-560067
(Autonomous Institution Affiliated to VTU, Belagavi)

**DEPARTMENT OF INFORMATION SCIENCE AND
ENGINEERING**

CERTIFICATE

Certified that the minor project work titled '*Personal expense tracker*' is carried out by **Akshay A Gowda (1MJ22IS009), Chethan SDN (1MJ22IS022), Dhiraj Acharya (1MJ22IS025) and Deekshith H A (1MJ22IS023)**, who are confide students of MVJ College of Engineering, Bengaluru, in partial fulfilment for the award of Degree of **Bachelor of Engineering in Information Science and Engineering** of the Visvesvaraya Technological University, Belagavi during the year 2024-2025. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the mini project report deposited in the departmental library. The minor project report has been approved as it satisfies the academic requirements in respect of mini project work prescribed by the institution for the said Degree.

Signature of Guide
Prof. Gayathri
Assistant Professor
Department of ISE
MVJCE
Bangalore -560067

Signature of HOD
Dr .Asha Joseph
Dean-School of CSE
Department of ISE
MVJCE
Bangalore-560067

Signature of Dean
Dr .Asha Joseph
Dean-School of CSE
MVJCE
Bangalore-560067

Name of Examiners
1.
2.

External Viva

Signature with Date

MVJ COLLEGE OF ENGINEERING, BENGALURU-560067
(Autonomous Institution Affiliated to VTU, Belagavi)

DEPARTMENT OF INFORMATION SCIENCE ENGINEERING

DECLARATION

We, Akshay A Gowda (1MJ22IS009), Chethan SDN (1MJ22IS022), Dhiraj Acharya (1MJ22IS025) and Deekshith H A (1MJ22IS023) students of fifth semester B.E., Department of Information Science and Engineering, MVJ College of Engineering, Bengaluru, hereby declare that the major project titled '*Personal expense tracker*' has been carried out by us and submitted in partial fulfilment for the award of Degree of **Bachelor of Engineering in Information Science and Engineering** during the year 2024-25

Further we declare that the content of the dissertation has not been submitted previously by anybody for the award of any Degree or Diploma to any other University.

We also declare that any Intellectual Property Rights generated out of this project carried out at MVJCE will be the property of MVJ College of Engineering, Bengaluru and we will be one of the authors of the same.

Place: Bengaluru

Date:

Name

Signature

- 1. Akshay A Gowda(1MJ22IS009)**
- 2. Chethan SDN (1MJ22IS022)**
- 3. Dhiraj Acharya(1MJ22IS025)**
- 4. Deekshith H A(1MJ22IS023)**

ACKNOWLEDGEMENT

We are indebted to our guide, **Prof Gayathri, Dept of Information Science and Engineering, MVJ College of Engineering** for his/her wholehearted support, suggestions and invaluable advice throughout our project work and also for the help in the preparation of this report.

Our sincere thanks to **Dr.Asha Joseph**, Dean-School of Computer Science and Engineering and Head, Department of Information Science and Engineering, MVJCE for her support and encouragement.

We express sincere gratitude to our beloved Principal, **Dr. Ajayan K R** for all his support towards this project work.

Lastly, we take this opportunity to thank our **family** members and **friends** who provided all the backup support throughout the project work.

ABSTRACT

- Managing personal finances effectively is essential for maintaining financial stability and achieving long-term goals. Our project, the Daily Expense Tracker, addresses this need by providing a user-friendly tool to monitor and categorize everyday expenditures. This application empowers users to track their spending patterns, helping them identify areas for saving and better budget management.
- The Daily Expense Tracker records expenses in real-time, categorizes them, and generates visual summaries, such as charts and graphs, to give users an overview of their spending habits. By simplifying financial monitoring and analysis, this project aims to promote responsible financial behavior and assist users in making informed financial decisions.

ACRONYMS

Acronym	Abbreviation
USN	University Seat Number
SEM	Semester

TABLE OF CONTENTS

Certificate		I
Declaration		II
Acknowledge		III
Abstract		IV
List Of Figures		VII
Chapter 1	Introduction	1-4
	1.1 Aim	1
	1.2 Motivation	2
	1.3 Problem Statement	3
	1.4 Existing System	3
	1.5 Proposed System	4
Chapter 2	Literature Survey	5-7
Chapter 3	System Requirements and Specification	8-12
	3.1 Hardware Requirements	9
	3.2 Software Requirements	9
	3.3 Functional and Non-Functional Requirements	10-12
Chapter 4	System Design	13-19
	4.1 System Architecture	14
	4.2 Data Flow Diagram	15-18
	4.3 Use Case Diagram	18-19

Chapter 5	Implementation	20-29
	5.1 Methodology	21
	5.2 Algorithm	22
	5.3 Code Snippet	23-29
Chapter 6	System Testing	30-34
	6.1 Unit Testing	31
	6.2 Integration Testing	32
	6.3 System Testing	32
	6.4 Test Cases	33-34
Chapter 7	Conclusion and Future Enhancement	35-41
References		
Appendix		

List of Figures

Figure No.	Figure Title	Page No.
4.1	System Architecture	19
4.2	Data Flow Diagram Level 0	22
4.3	Data Flow Diagram Level 1	23
4.4	Use Case Diagram	24
A.1	Screenshot of Home Page	48
A.2	Screenshot of Input Text	48
A.3	Screenshot of Summary Output	49

CHAPTER 1
INTRODUCTION

CHAPTER1

INTRODUCTION

In today's fast-paced world, keeping track of personal finances can be challenging. With expenses accumulating daily, it's easy to lose sight of spending habits, which can lead to overspending and financial stress. Many people struggle to find an effective way to monitor their spending patterns and often lack a clear understanding of where their money goes.

The Daily Expense Tracker project is designed to address these challenges. This application provides an easy-to-use platform for recording, categorizing, and analyzing expenses on a daily basis. By offering insights into spending habits, the tracker encourages mindful spending and helps users create a more balanced and sustainable budget.

Our solution not only aids in day-to-day expense tracking but also empowers users with data visualization and reporting tools, which facilitate better financial planning and decision-making. Through this project, we aim to simplify financial management.

1.1 AIM

The aim of the Personal Expense Tracker software project is to create a user-friendly, efficient tool that helps individuals manage their finances by tracking daily, weekly, and monthly expenses. The software will offer features such as expense categorization, budget setting, and real-time financial insights, empowering users to make informed decisions about their spending habits.

Additionally, the project will focus on intuitive design and security, ensuring that users can easily input data and access reports securely. The goal is to promote financial literacy and aid in personal budgeting for users of all ages.

The Personal Expense Tracker will also include features for setting financial goals, such as saving for specific purchases or building an emergency fund. With detailed visualizations like graphs and charts, users will be able to track progress and adjust their spending behaviors accordingly. The software will support multiple platforms, including mobile and desktop, allowing users to manage their finances anytime, anywhere. Through these features, the project aims to foster better financial habits, encourage savings, and provide greater control over personal finances.

1.2 MOTIVATION

The motivation behind the Personal Expense Tracker project stems from the increasing need for individuals to gain control over their finances in an increasingly complex and fast-paced world. Many people struggle with budgeting and tracking spending, leading to financial stress or missed opportunities for savings. By providing a simple yet powerful tool, this project aims to empower users to make smarter financial decisions, reduce debt, and build wealth. Ultimately, the goal is to help users develop healthier financial habits and achieve long-term financial stability.

1.3 PROBLEMSTATEMENT

Many individuals struggle with tracking their expenses, leading to poor financial decisions, overspending, and a lack of savings. Existing solutions may be complex or difficult to use, leaving people feeling overwhelmed. The problem this project addresses is the need for a simple, intuitive personal expense tracker that enables users to effectively manage their finances and achieve financial stability.

1.3.1 Problems Faced:

- Difficulty in tracking daily or monthly expenses.
- Lack of easy-to-use budgeting tools.
- Limited financial insights or actionable data.
- Overly complex or unintuitive financial software.
- Concerns over data security and privacy.
- Lack of motivation or accountability to stay on track.

1.3 ExistingSystem

Existing systems like **Mint**, **YNAB**, **Pocket Guard**, **Personal Capital**, **Spendee**, and **Good Budget** offer various features for tracking expenses, budgeting, and managing finances. However, they often struggle with complexity, lack of automation, or limited security. Each tool has strengths but may not fully meet the needs of all users.

1.3.1 MAJORLIMITATIONSARE:

- **Complexity and Usability**
- **Limited Automation**
- **Data Privacy and Security**
- **Inadequate Financial Insights**

1.4 ProposedSystem

The proposed system will also feature real-time syncing across devices, ensuring users can access their financial data anytime, anywhere. It will support multiple account integrations, such as bank accounts and credit cards, providing a comprehensive view of all financial transactions. Users will be able to generate detailed reports, including spending trends and budget comparisons, which will help them make informed decisions. Additionally, the system will include notifications and reminders to encourage users to stay on track with their financial goals and budgets.

The proposed system will incorporate advanced analytics to provide users with personalized financial advice based on their spending patterns and goals. Using machine learning algorithms, it will offer insights on potential savings opportunities and suggest areas to cut back on unnecessary expenses. The platform will also include a goal tracker, where users can set short-term and long-term financial objectives, such as building an emergency fund or saving for a major purchase, and receive tailored recommendations to achieve these goals more effectively.

CHAPTER-2

LITERATURE SURVEY

CHAPTER 2

LITERATURE SURVEY

The following research articles are selected for review, and discussed in detail.

[1]" Tracking the Expense using MERN Stack and Data Visualization" The central aim of the research was to identify an efficient approach to implement budgeting, delivering personalized budget plans based on individuals' income and spending preferences. The paper makes a noteworthy contribution by exploring links between prediction, categorization, and analysis of the expenses to implement tracking and budgeting.

[2]" Smart Personal Expense Tracker Technology” The objectives of a Smart Personal Expense Tracker Technology are multifaceted, aiming to address the diverse needs and challenges individuals face in managing their finances. Here are common objectives that such a Technology may strive to achieve:

[3]" PERSONAL EXPENSE TRACKER" Personal Expense Tracker (PET) is a daily expense management system which is specially designed for non- salaried and salaried personnel for keeping track of their daily expenditure with easy and effective way through computerized system which tends to eliminate manual paper works. It will also manage records in systematic way and user can access the stored data conveniently.

[4] " Case study on budget tracker app" Expense Tracker is going to be a cellular utility in order that it may be accessed any time required. This utility may have a -tier architecture: first one is the database tier, wherein all of the records and financial facts could be saved.

[5] “Data Analytics in personal Finance Management." The methodology followed for this research paper on "statistics Analytics in personal Finance management" is designed to investigate how individuals can leverage records analytics to optimize their private finance decisions, specializing in spending habits, savings, and investments.

[6] "An intelligent expense tracking system employing machine learning approaches" 2019 - The design and deployment of an intelligent expense monitoring system that employs machine learning methods to forecast future expenses based on historical data are described in this study. The authors discovered that their method accurately forecasts costs and enables consumers to manage their money more effectively.

[7] "Expensify: An expense tracker and report creator for mobile apps" (2018) authored by L. Zhang and Y. Wang Expensify is a mobile app based spending tracker and report maker, and this essay assesses its efficacy. According to the authors, Expensify is a helpful tool project title.

[8] "Personal Finance Management: The Design and Assessment of a Mobile Expense Tracker" by J. M. Clark et al. - An app called Cashew for tracking mobile expenses is designed and evaluated in this study. Cashew is a helpful tool, according to the developers, for managing personal finances while on the go and keeping track of expenditures

[9] "A survey of expense tracking applications" by A. M. Al-Naser and others (2021) - The popular expense-tracking applications Mint, Pocket Guard, and Personal Capital are reviewed in this essay. These apps, according to the authors, offer a variety of functions for managing personal money, such as budgeting, goal-setting, and investment tracking

[10] expense tracking," L. V. Ackerman and M. Weigel (2020). - In this study, the effectiveness of manually documenting spending in a notebook or spreadsheet vs automatically recording expenses is compared (i.e., using a software tool to automatically track expenses). The researchers discovered that automatic spending tracking is faster and more accurate than manual tracking.

CHAPTER 3

SYSTEM REQUIREMENTS AND SPECIFICATION

CHAPTER 3

SYSTEM REQUIREMENTS AND SPECIFICATION

3.1 HARDWARE REQUIREMENTS

- **Processor:** Intel Core i3 or equivalent AMD processor (minimum).
- **RAM:** 4 GB (8 GB or more recommended for smoother performance).
- **Storage:** 500 MB of available disk space for project files and dependencies.
- **Internet Connection:** Required for installing dependencies and using the gTTS service.

3.2 SOFTWARE REQUIREMENTS

1. **Operating System:** Windows 10/11, macOS, or Linux.
2. **Programming languages:** Java
3. **Java framework:** spring boot
4. **Web Browser:** Google Chrome, Mozilla Firefox, or any modern browser for testing and running the web application.
5. **Code Editor:** IntelliJ idea.

3.3 FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

3.3.1 Functional Requirements:

1. Expense Tracking:

The system must allow users to input, categorize, and track their daily, weekly, and monthly expenses automatically or manually.

2. Budget Management:

The system must enable users to set and manage budgets for different categories (e.g., groceries, entertainment, etc.) and track their progress in real time.

3. Bank and Account Integration:

The system should support integration with multiple bank accounts, credit cards, and payment platforms for automatic transaction importing.

4. Financial Insights and Reports:

Users should be able to generate reports and visualizations (e.g., graphs, pie charts) to view spending patterns, trends, and budget performance.

5. Goal Setting and Tracking:

The system should allow users to set financial goals (e.g., saving for a vacation, paying off debt) and track progress toward these goals.

6. Notifications and Alerts:

Users should receive notifications for budget limits, bill due dates, or when financial goals are met or missed.

7. Expense Categorization:

The system should automatically categorize transactions (e.g., groceries, entertainment) and allow users to modify categories when necessary.

8. Multi-Device Syncing:

Data should sync in real time across multiple devices (mobile, desktop) to ensure users always have access to the most up-to-date financial information

3.4.2 Non-Functional Requirements:**1. Performance:**

- The application should process text summarization requests efficiently, with a response time of under 3 seconds for typical inputs.
- The application should support multiple users simultaneously without significant degradation in performance.

2. Scalability:

- The system should be designed to handle increasing numbers of text inputs without requiring significant changes to the underlying architecture.

3. Usability:

- The web interface should be intuitive, easy to navigate, and user-friendly for both technical and non-technical users.
- The design should be responsive, ensuring compatibility with various screen sizes, including desktops, tablets, and mobile devices.

4. Reliability:

- The application should be available with minimal downtime, ensuring users can access the service whenever needed.
- The system should be robust, with a high success rate for generating summaries and converting text to speech.

5. Maintainability:

- The codebase should be modular and well-documented, making it easy to maintain, update, and extend.
- The system should be designed to accommodate future enhancements, such as additional languages or summarization techniques.

CHAPTER-4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

The system architecture for the Personal Expense Tracker is based on a client-server model. The frontend, consisting of a mobile app and web interface, enables users to track expenses, set budgets, and view financial reports. The backend handles data processing, expense management, and goal tracking, integrating with third-party APIs for automatic transaction imports. User data is securely stored in a relational or NoSQL database with encryption for privacy. Real-time syncing across devices ensures seamless access to data, while notifications alert users to budget limits, transactions, and financial milestones. The architecture is scalable, secure, and designed for easy user interaction and data protection.

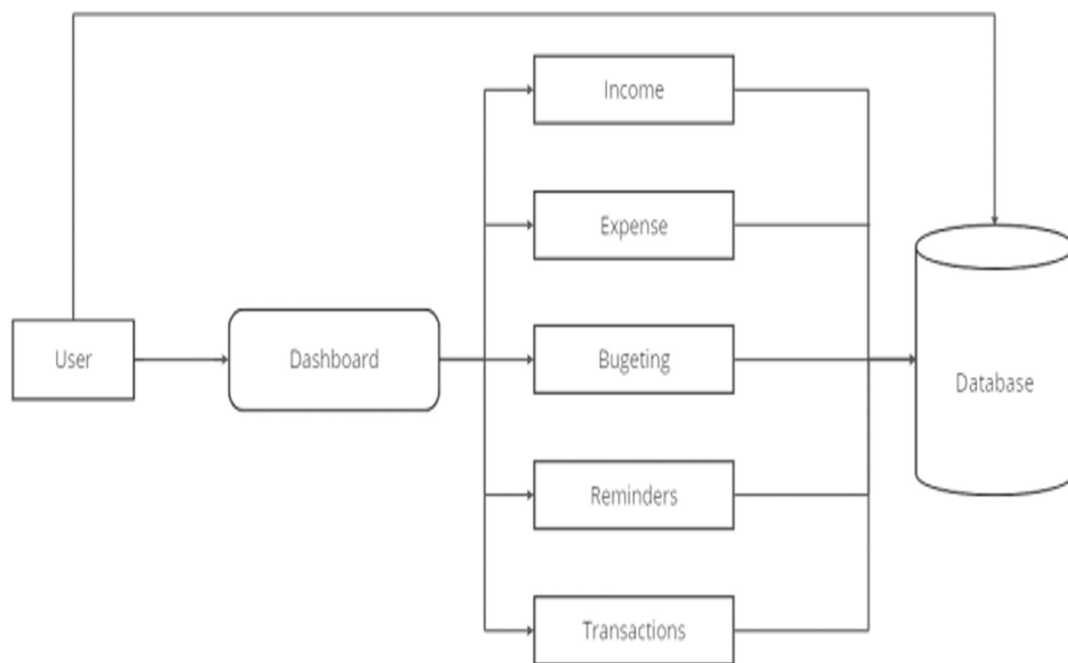


Fig. 4.1: System Architecture

The Personal Expense Tracker system architecture is designed around a modular client-server model. The frontend, accessible via mobile and web applications, provides an intuitive user interface for tracking expenses, managing budgets, and setting financial goals. The backend processes user data through services like expense categorization, budgeting, and report generation. It integrates with third-party financial APIs for automatic transaction imports from bank accounts and payment platforms. User data is securely stored in a database, with encryption ensuring privacy. Real-time syncing across devices ensures seamless access, while notifications keep users informed about budget limits and financial goals. The architecture ensures scalability, security, and performance.

4.2 DATA FLOW DIAGRAM

The Data Flow Diagrams (DFDs) for the Text Summarization using NLP with Voice Summary project depict the flow of data across three levels of abstraction. At Level 0, the system is viewed as a single process where the user inputs text, and the system outputs a summary and voice summary. Level 1 breaks down the system into its core components: input text is pre-processed, passed through a summarization process using a text-ranking algorithm, and then converted to a summary and voice output. This level details the interaction between user input and system processes. At Level 2, the preprocessing stage is further decomposed to show steps like removing stop words and cleaning the text. The summarization process is also detailed, illustrating the application of the text-ranking algorithm, followed by the generation of a summary, which is then converted into a voice output. Each level provides a deeper understanding of the system's processes, interactions, and data transformations from user input to final output.

4.2.1 Data Flow Diagram Components

Level 0: High-Level Overview

At **Level 0**, the diagram shows the primary system's interaction with external entities, representing the system's major processes and data flow.

Entities:

1. **User** – Interacts with the system, inputs data, and views reports.
2. **Third-Party Financial APIs** – Provide transaction data (e.g., bank, credit card APIs).

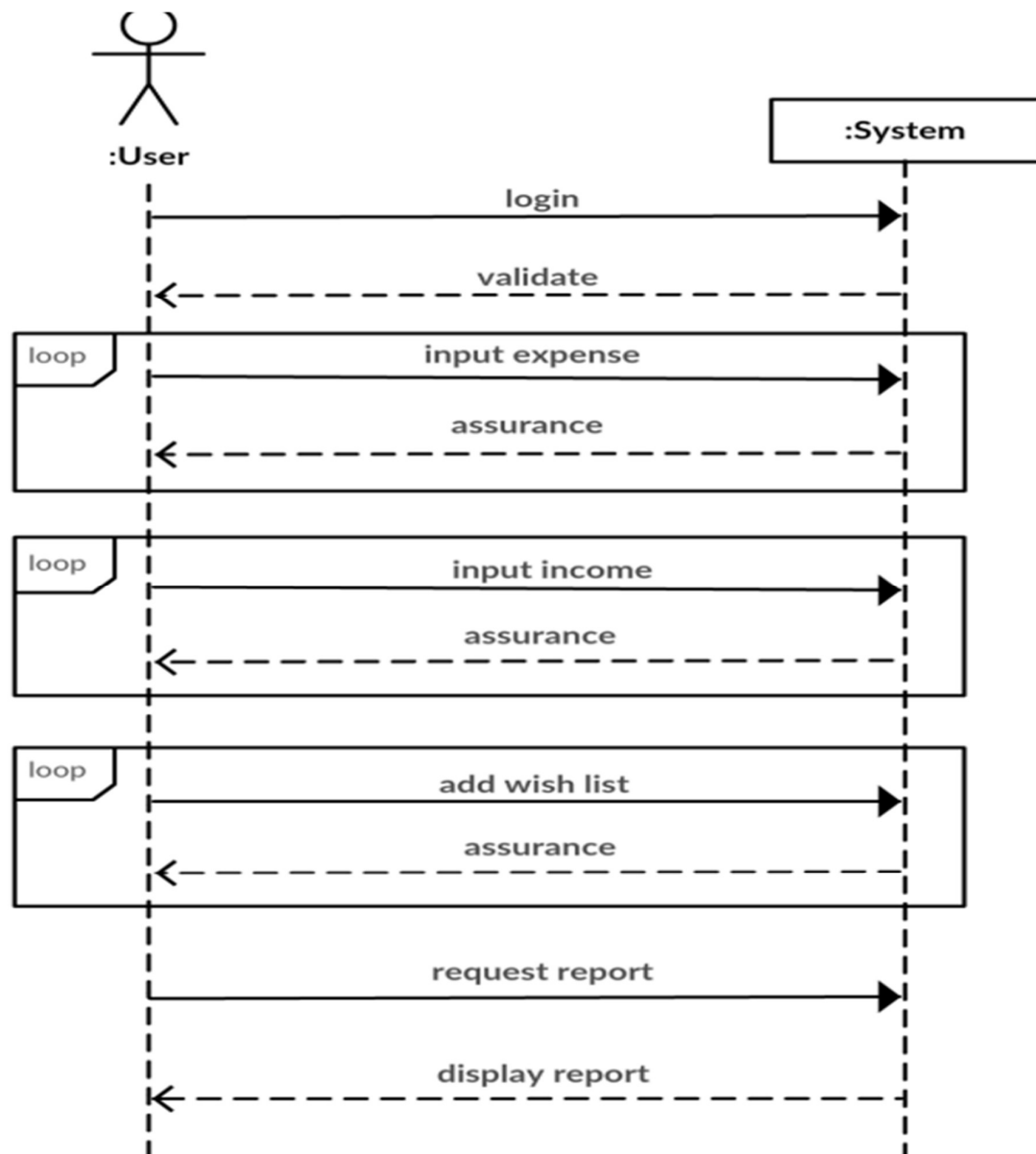
Processes:

1. **Expense Tracker System** – The central system where all data is processed.

Data Stores:

1. **User Data Store** – Stores user credentials, preferences, and historical financial data.
2. **Expense Data Store** – Stores expense details, categories, and transaction records.
3. **Budget Data Store** – Stores user-set budgets and financial goals.

Data Flow - The movement of data is shown by pointed arrows. Data development is shown from the bottom of the arrow, which represents its source, to the top of the arrow, which represents its destination.

**Fig. 4.2: Data Flow Diagram Level 0**

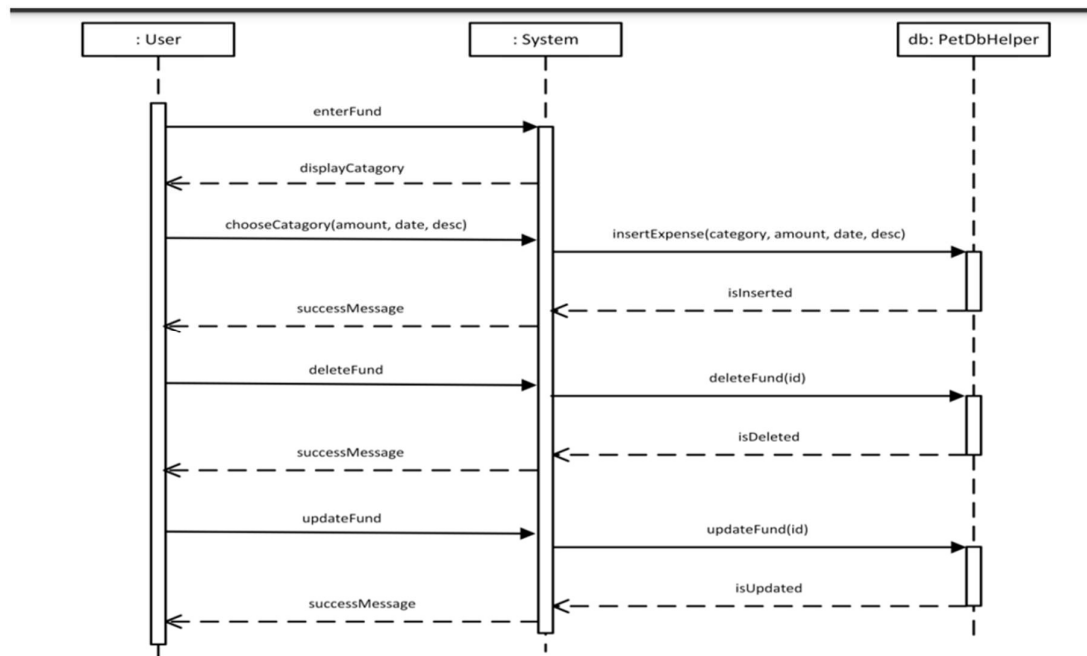


Fig. 4.3: Data Flow Diagram Level 1

4.4 USE CASE DIAGRAM

The Use Case Diagram for the Personal Expense Tracker outlines the interaction between users and the system. The primary actor, the User, engages with various functionalities, including Login/Authentication, Track Expenses, Categorize Expenses, Set Budget, View Financial Reports, and Set Financial Goals. The system also integrates with Bank/Financial APIs to automatically Sync Transactions, updating user expense records in real-time. Notifications are sent to users for budget limits, upcoming bills, and financial milestones. This structure highlights how the system helps users manage their personal finances by tracking spending, setting budgets, achieving financial goals, and generating reports.

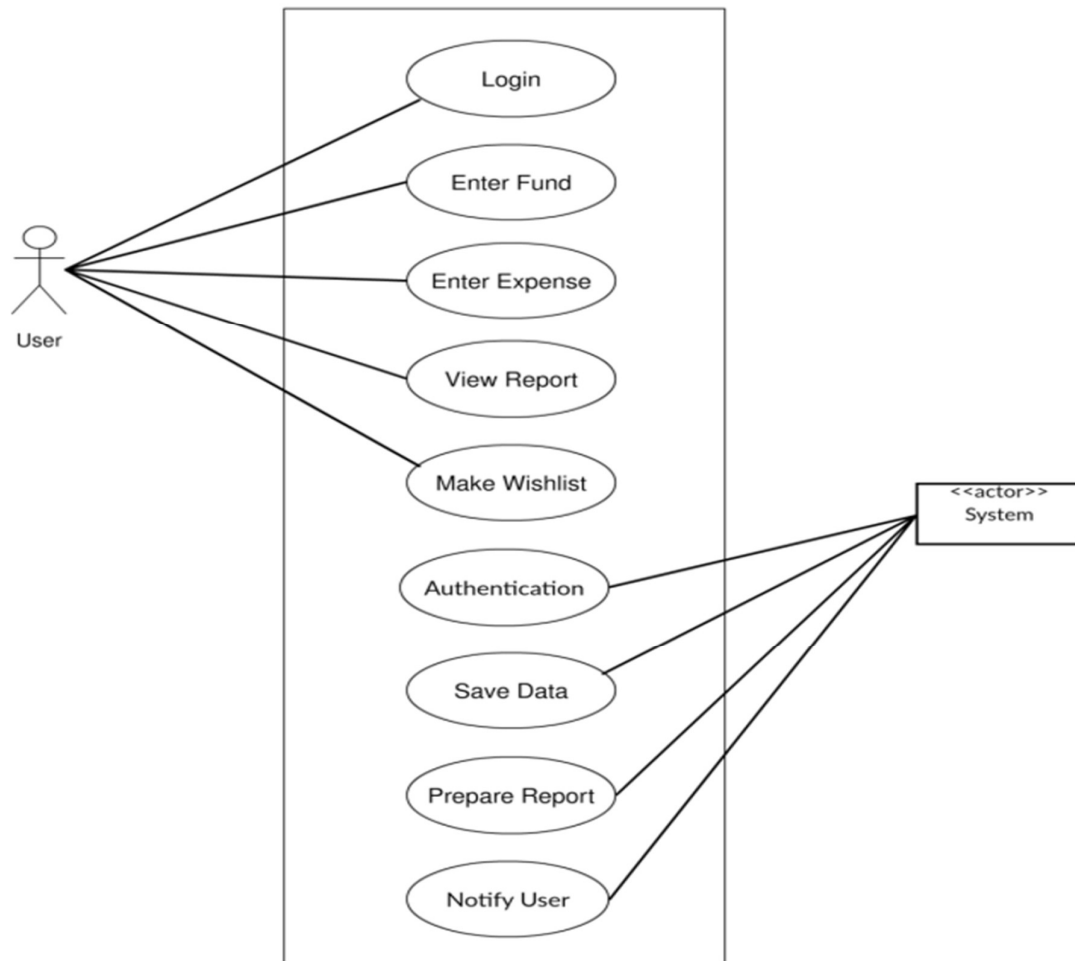


Fig. 4.4: Use Case Diagram for User

CHAPTER-5

IMPLEMENTATION

CHAPTER5

IMPLEMENTATION

5.1 METHODOLOGY

Develop a personal expense tracker by defining objectives, selecting a tracking tool (e.g., app or spreadsheet), and creating income and expense categories. Record transactions daily, organize data by category, and analyze trends monthly. Set budgets and goals based on insights to optimize spending, improve savings, and achieve financial stability.

5.1.1 HTML

HTML (Hyper Text Markup Language) is the standard language for creating and designing web pages. It defines the structure and content of a web page using various tags and elements.

In your project, `index.html` serves as the home page of the application where users input text for summarization.

- **Structure:** The HTML document starts with the `<!DOCTYPE html>` declaration, followed by the `<html>` tag that contains the entire document. Inside, there are `<head>` and `<body>` sections.
- **Metadata:** The `<head>` section includes meta tags for character encoding, compatibility, and viewport settings. It also links to external CSS stylesheets and Bootstrap for styling.

When a user visits the home page, `index.html` is rendered by Flask. It displays a text area for input and a button to submit the text. The form data is sent to the `/analyze` route when submitted.

5.1.2

CSS (Cascading Style Sheets) is used to control the presentation and layout of HTML documents. It defines how HTML elements should be styled, including colors, fonts, spacing, and positioning.

The style.css file customizes the appearance of the web pages (index.html and summary.html) to ensure a consistent and attractive user interface.

- **Body Styles:** The body selector applies a default font family (Arial, sans-serif) to the

ALGORITHM

1. Data Organization and Storage

CRUD Operations: For managing user data in databases (Create, Read, Update, Delete).

Sorting Algorithms: To display transactions chronologically or by amount (e.g., Quick Sort, Merge Sort).

Search Algorithms: For finding specific transactions or categories (e.g., Binary Search, Linear Search).

2. Categorization

Rule-Based Categorization: Simple matching algorithms to assign categories based on keywords in transaction descriptions.

Machine Learning Models : Naïve Bayes: For text classification of transaction descriptions .k-Nearest Neighbors (k-NN): For categorizing transactions based on similarity to previously categorized transactions.

5.2 CODESNIPPET

5.2.1 index.html File:

```
- <!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
  <title>Error</title>
  <link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.min.css"
rel="stylesheet" integrity="sha384-
EVSTQN3/azprG1Anm3QDgpJLIm9Nao0Yz1ztcQTwFspd3yD65VohhpuuCOmLA
SjC" crossorigin="anonymous">
  <link rel="stylesheet" th:href="@{/css/errorPageStyle.css}" />
</head>
<body>
<div class="page-container">

  <div class="image-container">
    
  </div>
  <div class="text-container">
    <div class="big-text-container" th:unless="{errorMessage}">
```

```
<h1 style="text-align: center">404 Not Found</h1>
</div>
<div class="big-text-container" th:if="{errorMessage}">
  <h1 th:text="{errorMessage}" style="text-align: center"></h1>
</div>
<div class="small-text-container">
  <h4>The content you are looking for might have been removed, had its name
changed or is temporarily unavailable.</h4>
</div>
<div class="button-container">
  <a th:href="@{/expenses}">
    <button class="btn btn-primary">Go to Expenses</button>
  </a>
</div>
</div>
<div class="empty-container">

</div>
</div>
</body>
</html>
```

5.2.2 summary.html File:

```
<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
  <meta charset="UTF-8">
  <title>New expense type</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.min.css"
rel="stylesheet" integrity="sha384-
EVSTQN3/azprG1Anm3QDgpJLIm9Nao0Yz1ztcQTwFspd3yD65VohhpuuCOmLASjC"
crossorigin="anonymous">
  <link rel="stylesheet" th:href="@{/css/styles.css}" />
  <link rel="preconnect" href="https://fonts.googleapis.com">
  <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin>
  <link href="https://fonts.googleapis.com/css2?family=Fira+Code&display=swap"
rel="stylesheet">
  <style>
    body {
      font-family: 'Fira Code', monospace;
    }
  </style>

</head>
<body class="body">
<nav class="navbar navbar-expand-lg">
  <div class="container-fluid">
    <a class="navbar-brand" style="color: #232526" th:href="@{/expenses}">Expense
Tracker System</a>
    <button class="navbar-toggler" type="button" data-bs-toggle="collapse" data-bs-
target="#navbarText" aria-controls="navbarText" aria-expanded="false" aria-
label="Toggle navigation">
```

```

    <span class="navbar-toggler-icon"></span>
  </button>
</div>
</nav>

<div class="container-fluid" id="container" >
  <div class="center-column">
    <div><br></div>

    <div class="container-sm">
      <h3 class="tableHead">Types of expense</h3>
      <table class="table table-hover" id="data-table">
        <thead>
          <tr>
            <th scope="col">Type of expense</th>
            <th scope="col">Delete</th>
          </tr>
        </thead>
        <tbody>
          <tr th:each="expenseType : ${expenseTypes}">
            <td th:text="${expenseType.expenseCategory}"></td>
            <td>
              <form th:action="@{/newExpenseType/delete/{id}}(id=${expenseType.id})"
th:method="post">
                <input width="25" height="25" type="image" src="https://img.icons8.com/ios-
glyphs/30/filled-trash.png" alt="filled-trash" />
              </form>
            </td>
          </tr>
        </tbody>
      </table>

```

```

    </div>
  </div>
  <div class="add-category-column">
    <div><br></div>
    <div><br></div>
    <div class="newExpenseContent">
      <h3 class="tableHead">Add new expense type</h3>
      <br>
      <form th:action="@{/newExpenseType}" th:object="{expenseType}"
method="post">

        <div class="mb-3"> <!--name of expense-->
          <label for="name" class="form-label">New expense type</label>
          <input type="text" id="name" class="form-control form-rounded"
th:field="*{expenseCategory}" th:errorclass="is-invalid" aria-
describedby="nameHelp">
            <div id="validationName" class="invalid-feedback"
th:errors="*{expenseCategory}">
              </div>
            </div>
            <div th:if="{errorMessage}" class="alert alert-danger">
              <p th:text="{errorMessage}"></p>
            </div>

            <br>
            <button type="submit" class="btn btn-outline-dark btn-sm btn-
shadow">Add</button>

            <a th:href="@{/expenses}" class="btn btn-dark btn-sm btn-shadow cancel-btn">Go
back</a>
          </form>

```

```
</div>
<!-- ----->
</div>
</div>
</body>
</html>
```

```

<!DOCTYPE html>
<html xmlns:th="http://www.thymeleaf.org">
<head>
  <meta charset="UTF-8">
  <title>Update Expense</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.0.2/dist/css/bootstrap.min.css"
rel="stylesheet" integrity="sha384-
EVSTQN3/azprG1Anm3QDgpJLl9Nao0Yz1ztcQTwFspd3yD65VohhpuuCOmLASj
C" crossorigin="anonymous">
  <link rel="stylesheet" th:href="@{/css/updateStyle.css}" />

</head>
<body class="body">

<div class="container-fluid" id="container" >

  <div class="center-column">
    <div><br></div>
    <div><br></div>
    <!-- Sidebar content -->
    <div class="updateCont">
      <h3 class="tableHead">Edit your expense:</h3>
      <form th:action="@{/update}" th:object="${expense}" method="post"
novalidate>
        <div class="mb-3"> <!--id of expense-->
          <input type="hidden" id="id" class="form-control" th:field="*{id}"
th:errorclass="is-invalid" aria-describedby="idHelp">
          <div id="validationId" class="invalid-feedback" th:errors="*{id}">
            </div>
        </div>

        <div class="mb-3"> <!--name of expense-->
          <label for="name" class="form-label">Name of expense</label>
          <input type="text" id="name" class="form-control" th:field="*{name}"
th:errorclass="is-invalid" aria-describedby="nameHelp">
          <div id="validationName" class="invalid-feedback" th:errors="*{name}">
            </div>
        </div>

        <!--type of expense-->
        <div class="mb-3">

```

CHAPTER-6

SYSTEM TESTING

CHAPTER 6

SYSTEM TESTING

System testing in a personal expense tracker is a vital process to ensure the application operates seamlessly and fulfills user expectations. It involves evaluating the entire system to verify that all features, from adding transactions and categorizing expenses to generating budgets and reports, function as intended. This phase assesses the application's reliability, performance under various conditions, and security of sensitive financial data. By simulating real-world scenarios, system testing ensures the tracker is user-friendly, compatible across devices, and integrates smoothly with external services like bank APIs or cloud storage. The primary objective is to identify and resolve any defects, ensuring that users can manage their finances accurately and efficiently without disruptions.

6.1 UNIT TESTING

Unit testing for a personal expense tracker ensures that individual components function correctly in isolation. Each module, such as adding transactions, calculating totals, or categorizing expenses, is tested to verify accuracy and reliability. For instance, tests are conducted to check if a newly added expense updates the total correctly and if category filters display relevant data. These tests are typically automated, using tools like JUnit or pytest, to ensure consistency and repeatability. Edge cases, such as negative values or empty fields, are also examined to ensure robust error handling. By identifying bugs early, unit testing helps maintain the overall quality of the application and ensures that core functionalities perform as intended before integration with other components.

6.2INTEGRATIONTESTING

Integration testing for a personal expense tracker focuses on verifying the interaction between different modules to ensure they work seamlessly as a system. This testing ensures that core features, such as adding transactions, updating totals, categorizing expenses, and generating reports, function cohesively when integrated. For example, when a user adds a transaction, the system should update the total in the corresponding category, reflect changes in the budget, and adjust visual charts accordingly. Additionally, integration testing checks connections with external services like bank APIs for importing transactions, payment gateways for expense tracking, or cloud storage for data backup. Edge cases, such as data mismatches or synchronization delays, are tested to ensure the application handles errors gracefully. By thoroughly testing interdependencies, integration testing guarantees a smooth and reliable user experience.

6.3SYSTEM TESTING

System testing is a critical phase in evaluating the functionality, reliability, and performance of a personal expense tracker application. It ensures that the system operates as intended and meets user requirements under various conditions. Functional aspects such as adding, editing, and deleting transactions are rigorously tested to verify accuracy in calculations and categorization of expenses. Usability is assessed by analyzing the application's interface for intuitiveness, ease of navigation, and accessibility across devices like desktops, tablets, and mobile phones. Performance testing focuses on the application's responsiveness, ensuring it can handle large datasets or concurrent users without lag. Security testing is equally important, emphasizing the protection of sensitive financial data through encryption and safeguarding against unauthorized access. Compatibility testing ensures the application functions seamlessly across different operating systems and browsers, while integration testing verifies smooth synchronization with third-party tools such as banking APIs or cloud services. Finally, regression testing ensures that updates or bug fixes do not disrupt existing features, maintaining the application's overall stability.

6.3 TEST CASES

Test Case#	TC01
Test Name	Addition of field
Test Description	Verify adding a valid expense with all required fields.
Input	Click "Add Expense", enter Date (today), Amount (10.50), Category (Food), Description (Lunch), click "Save"
Expected Output	Expense is added to the list with correct details.
Actual Output	Total expenses generated.
Test Result	Success

Table 6.4.1 Test case for Addition of field

Test Case#	TC02
Test Name	Edit the field
Test Description	Verify editing an existing expense's amount.
Input	Select an existing expense, change the Amount to 20.00, click "Save".
Expected Output	The selected expense's amount is updated to 20.00
Actual Output	Expense is updated.
Test Result	Success

Table 6.4.2 Test case for Edit the field

Test Case#	TC03
Test Name	View Total
Test Description	Verify viewing total expenses for the current month.
Input	Select the current month in the date range filter, click "View Total".
Expected Output	The total expenses for the current month are displayed correctly.
Actual Output	Expense of current month.
Test Result	Success

Table 6.4.3 Test case for View Total

Test Case#	TC04
Test Name	Search for field
Test Description	Verify searching for expenses by category
Input	Enter "Food" in the search bar, click "Search".
Expected Output	All expenses categorized as "Food" are displayed
Actual Output	Expense spent on food.
Test Result	Success

Table 6.4.4 Test case for Search field

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENT

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENT

In conclusion, a personal expense tracker is an essential tool for managing finances effectively, promoting better budgeting, saving habits, and financial awareness. Through features like transaction tracking, categorization, and insightful reports, users can make informed decisions about their spending. Ensuring the application is thoroughly tested for reliability, usability, and security enhances user confidence and satisfaction.

Future enhancements could include advanced features like AI-driven spending insights, predictive analytics for budgeting, and integration with multiple financial platforms. Adding voice-enabled data entry and cross-device synchronization could improve accessibility. Enhanced security measures like biometric authentication and blockchain for data integrity would address user concerns about privacy. Such innovations will make the tracker more robust, personalized, and aligned with evolving financial management needs.

FUTURE ENHANCEMENT

Future enhancements for a personal expense tracker could include AI-powered insights to analyze spending patterns and offer personalized budgeting recommendations. Integration with multiple financial institutions and automatic transaction imports from various sources can streamline tracking. Adding features like voice-activated expense input, real-time notifications for budget limits, and predictive analytics for future expenses would improve user experience. Additionally, implementing stronger security features like biometric authentication and utilizing blockchain for data transparency and integrity could boost trust. These improvements would make the tracker more intelligent, user-friendly, and secure for managing finances.

REFERENCES

[1]" Tracking the Expense using MERN Stack and Data Visualization" The central aim of the research was to identify an efficient approach to implement budgeting, delivering personalized budget plans based on individuals' income and spending preferences. The paper makes a noteworthy contribution by exploring links between prediction, categorization, and analysis of the expenses to implement tracking and budgeting.

[2]" Smart Personal Expense Tracker Technology” The objectives of a Smart Personal Expense Tracker Technology are multifaceted, aiming to address the diverse needs and challenges individuals face in managing their finances. Here are common objectives that such a Technology may strive to achieve:

[3]" PERSONAL EXPENSE TRACKER" Personal Expense Tracker (PET) is a daily expense management system which is specially designed for non- salaried and salaried personnel for keeping track of their daily expenditure with easy and effective way through computerized system which tends to eliminate manual paper works. It will also manage records in systematic way and user can access the stored data conveniently.

[4] " Case study on budget tracker app" Expense Tracker is going to be a cellular utility in order that it may be accessed any time required. This utility may have a -tier architecture: first one is the database tier, wherein all of the records and financial facts could be saved.

[5] “Data Analytics in personal Finance Management." The methodology followed for this research paper on "statistics Analytics in personal Finance management" is designed to investigate how individuals can leverage records analytics to optimize their private finance decisions, specializing in spending habits, savings, and investments.

[6] "An intelligent expense tracking system employing machine learning approaches" 2019 - The design and deployment of an intelligent expense monitoring system that employs machine learning methods to forecast future expenses based on historical data are described in this study. The authors discovered that their method accurately forecasts costs and enables consumers to manage their money more effectively.

[7] "Expensify: An expense tracker and report creator for mobile apps" (2018) authored by L. Zhang and Y. Wang Expensify is a mobile app based spending tracker and report maker, and this essay assesses its efficacy. According to the authors, Expensify is a helpful tool for both people and organisations to manage their costs and produce reports for tax and reimbursement purposes.

[8] "Personal Finance Management: The Design and Assessment of a Mobile Expense Tracker" by J. M. Clark et al. - An app called Cashew for tracking mobile expenses is designed and evaluated in this study. Cashew is a helpful tool, according to the developers, for managing personal finances while on the go and keeping track of expenditures

[9] "A survey of expense tracking applications" by A. M. Al-Naser and others (2021) - The popular expense-tracking applications Mint, Pocket Guard, and Personal Capital are reviewed in this essay. These apps, according to the authors, offer a variety of functions for managing personal money, such as budgeting, goal-setting, and investment tracking

[10] expense tracking," L. V. Ackerman and M. Weigel (2020). - In this study, the effectiveness of manually documenting spending in a notebook or spreadsheet vs automatically recording expenses is compared (i.e., using a software tool to automatically track expenses). The researchers discovered that automatic spending tracking is faster and more accurate than manual tracking.

APENDIX A: Snapshots

Expense Tracker System

Filter

Select Year:
2023

Select Month:
Select the mor

Type of expense:
Choose the typ

Filter

Show all

Total Expenses: 0₹

No expenses found

Add expense below:

Name of expense

Type of expense
Choose the type of expense

Customize your types of expense

Amount

Date of birth
dd-mm-yyyy

+ Add expense

Fig.A.1ScreenshotofHomePage.

Filter

Select Year:
2023

Select Month:
Select the mor

Type of expense:
Choose the typ

Filter

Show all

Total Expenses: 1150.00₹

All expenses

Name of expense	Type of expense	Amount	Date	Update	Delete
books	education	100.00₹	2024-12-04		
fuel	travel	500.00₹	2024-12-03		
ticket	entertainment	500.00₹	2024-12-04		
books	education	50.00₹	2024-12-11		

Previous **1** Next

Download as CSV

Add expense below:

Name of expense

Please specify the name of expense

Type of expense
Choose the type of expens

Please specify the name of expense

Customize your types of expense

Amount

Please specify an amount

Date of birth
dd-mm-yyyy

Date cannot be empty!

+ Add expense

Fig. A.2Screenshot of Input Text.

Expense Tracker System

Types of expense

Type of expense	Delete
education	
entertainment	
food	
Home	
travel	

Add new expense type

New expense type

Add

Go back

Fig.A.3ScreenshotofSummary Output.